Homework #1

Due: 23:59pm October 15, 2026

Problem

Train a simple linear regression model to predict house prices as a function of their floor s ize, by running gradient descent for 500 epochs with a learning rate of 0.1. Use the dataset provided (train.txt and test.txt).

- Plot J(w) vs. the number of epochs.
- After training print the parameters and compare with solution from normal equations.
- Show your results using plot by ploting the train data using the default blue circles and test data using green xs. On the same graph show the linear solution line.

NOTE: Implement your solution on skeleton files that are provided. (simple_regression.py and scaling.py)